

CVaR Portfolio Optimization: A Risk Management Analysis

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Abstract

This study examines five portfolio optimization strategies using Conditional Value-at-Risk (CVaR) minimization across different market conditions. We analyze standard CVaR optimization, sensitivity across confidence levels, minimax approaches, dynamic monthly rebalancing, and stability-constrained implementations. Stability constraints successfully maintain most risk bene-fits while enforcing practical implementation limits. The findings demonstrate that CVaR frameworks can effectively adapt to different objectives, from average performance to extreme tail protection, with distinct risk-return-stability trade-offs for varying investor preferences.

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1. Data Preparation and Return Calculations

Task 1 involves the fundamental data preprocessing steps required for portfolio optimization. We calculate daily percentage returns from stock price data for both 2019 and 2020 periods using the formula: $\text{returns} = (\text{price}_t - \text{price}_{t-1}) / \text{price}_{t-1}$. This transformation converts absolute price movements into comparable percentage changes, enabling meaningful comparison across different assets and providing the foundation for all subsequent risk calculations. The data preparation includes handling missing values, ensuring consistent time periods, and separating the NASDAQ index (NDX) from individual stock returns to use as a benchmark.

2. CVaR Portfolio Optimization

The CVaR optimization problem requires careful design of decision variables, constraints, and the objective function to transform the theoretical framework into a tractable linear program.

```
# Decision variables
optx = optMod.addMVar(shape=n_stocks, lb=0.0)           # portfolio weights x_j
alpha = optMod.addVar(lb=-gp.GRB.INFINITY)             # VaR threshold
u = optMod.addMVar(shape=q, lb=0)                     # auxiliary variables u_k
```

Listing 1: Decision Variables

The optimization model employs three sets of decision variables:

1. **Portfolio Weights ($\mathbf{x}$):** Vector of n variables representing the proportion invested in each asset. Lower bound of 0.0 enforces the no short-selling constraint.
2. **VaR Threshold (α):** Scalar variable representing the Value-at-Risk threshold. Unrestricted bounds allow for negative values, which represent gains rather than losses.
3. **Auxiliary Variables ($\mathbf{u}$):** Vector of q variables (where q is the number of scenarios/ time periods) that capture tail losses exceeding the VaR threshold. These variables linearize the $[\cdot]^+$ function in the CVaR formula.

```
# Constraint 1: Budget constraint
optMod.addConstr(optx.sum() == 1)

# Constraint 2: Non-negativity (no short selling) # Handled by lb
# =0.0 in variable definition

# Constraint 3: Tail loss capture
optMod.addConstr(u >= -Y @ optx - alpha)
```

```
# Constraint 4: Auxiliary variable non-negativity # Handled by
lb=0 in u variable definition
```

```
# Constraint 5: Minimum return requirement
optMod.addConstr(mean_returns @ optx >= R_target)
```

The model incorporates five essential constraint sets to ensure a valid and meaningful portfolio:

Constraint	Description
Budget Constraint	$\sum_{j=1}^n x_j = 1$ Ensures portfolio weights sum to exactly 1.0, representing full investment of available capital without leverage.
Non-Negativity	$x_j \geq 0 \quad \forall j = 1, \dots, n$ Prevents short positions by requiring all portfolio weights to be non-negative.
Tail Loss Capture	$u_k \geq -\mathbf{x}^T \mathbf{y}_k - \alpha \quad \forall k = 1, \dots, q$ Linearization constraint: each u_k captures the excess loss beyond the VaR threshold. Effectively implements $u_k = \max(0, -\mathbf{x}^T \mathbf{y}_k - \alpha)$.
Auxiliary Non-Negativity	$u_k \geq 0 \quad \forall k = 1, \dots, q$ Ensures auxiliary variables remain non-negative, completing the linearization of $[\cdot]^+$.
Minimum Return Requirement	$\mathbf{x}^T \bar{\mathbf{y}} \geq R$ Ensures expected portfolio return exceeds the target threshold $R = 0.0002$ (0.02% daily).

Table 1: Summary of CVaR Portfolio Optimization Constraints

The objective function directly implements the CVaR formula:

```
optMod.setObjective(alpha + (1/((1-beta)*q)) * u.sum(), gp.GRB.MINIMIZE)
```

This formulation minimizes:

$$\min_{\mathbf{x}, \alpha, \mathbf{u}} \quad \alpha + \frac{1}{(1-\beta)q} \sum_{k=1}^q u_k$$

where:

- α : The VaR threshold
- $(1-\beta)^{-1}$: Scaling factor (20 for $\beta = 0.95$) that amplifies tail risk importance

$$\frac{1}{q} \sum_{k=1}^q u_k: \text{Average tail loss beyond VaR}$$

Out-of-Sample Evaluation

For testing portfolio performance on new data, we avoid re-optimization and directly calculate CVaR:

```
def calculate_cvar(returns, weights, beta):
    portfolio_returns = np.dot(returns.values, weights)
    portfolio_losses = -portfolio_returns
    var_val = np.percentile(portfolio_losses, beta * 100)
    tail_losses = portfolio_losses[portfolio_losses >= var_val]
    cvar_val = np.mean(tail_losses) if len(tail_losses) > 0 else var_val
    return cvar_val, var_val
```

The optimization results demonstrate significant performance differences between in-sample and out-of-sample periods:

- **In-sample CVaR (2019):** 0.0111 (1.11%)
- **Out-of-sample CVaR (2020):** 0.0458 (4.58%)
- **CVaR deterioration:** 313.3%

Benchmark Comparison:

- **NDX CVaR 2019:** 0.0244, **2020:** 0.0559
- **NDX deterioration:** 128.9%

The substantial CVaR deterioration (313.3% vs. 128.9% for NDX) indicates severe non-stationarity in return distributions between the normal market conditions of 2019 and the COVID-19 disrupted markets of 2020. However, the optimized portfolio maintained superior risk characteristics relative to the benchmark (CVaR: 0.0458 vs. 0.0559), demonstrating CVaR optimization effectiveness even under extreme market stress.

The results highlight the limitations of static allocation strategies while validating the CVaR framework's ability to construct portfolios with superior tail risk properties. The 313.3% deterioration emphasizes the critical importance of dynamic risk management that adapts to changing market conditions, suggesting that monthly or quarterly rebalancing may be necessary for maintaining optimal risk profiles.

Figure 2 illustrates the portfolio loss distributions for 2019 and 2020, clearly showing the dramatic shift in tail risk characteristics between the stable market conditions of 2019 (CVaR: 0.0110) and the volatile COVID-19 period of 2020 (CVaR: 0.0458), with the 2020 distribution exhibiting significantly heavier left tails and more extreme loss scenarios.

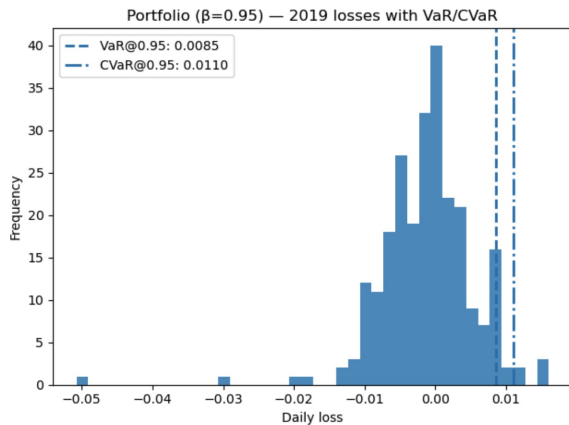


Figure 1: Portfolio losses with VaR/CVaR for 2019

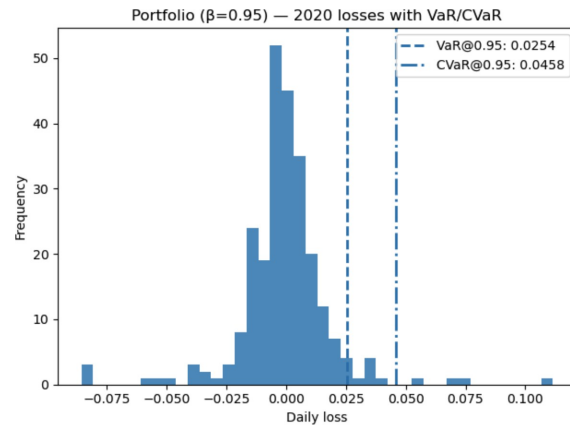


Figure 2: Portfolio losses with VaR/CVaR for 2020

3. Impact of Beta Values on Portfolio Allocation

In this section, we examine how different confidence levels (β) affect portfolio composition and risk characteristics. We optimize portfolios using 2019 data with three different beta values: 0.90, 0.95, and 0.99, representing 90%, 95%, and 99% confidence levels respectively. Each portfolio is then evaluated on both in-sample (2019) and out-of-sample (2020) data to assess robustness across market regimes.

The confidence level β determines which tail scenarios the CVaR optimization targets:

- $\beta = 0.90$: Focuses on the worst 10% of scenarios
- $\beta = 0.95$: Focuses on the worst 5% of scenarios
- $\beta = 0.99$: Focuses on the worst 1% of scenarios

Table 2: Comparative Analysis Across Beta Values

Beta	CVaR 2019	CVaR 2020	Stocks Used	Max Weight	Top 5 Conc.	Portfolio Return
0.90	0.0089	0.0321	16	0.2406	0.7073	0.0012
0.95	0.0111	0.0458	13	0.3039	0.7780	0.0012
0.99	0.0125	0.0909	9	0.4470	0.9024	0.0013

Table 2 presents the comparative analysis across different beta values. The results reveal a clear relationship between beta values and portfolio concentration:

1. Diversification Effect:

- $\beta = 0.90$: Uses 16 stocks with maximum weight of 24.06%
- $\beta = 0.95$: Concentrates to 13 stocks with maximum weight of 30.39%

- $\beta = 0.99$: Highly concentrated with only 9 stocks and maximum weight of 44.70%

2. **Weight Concentration Intensifies:** The maximum individual weight increases dramatically from 24.06% ($\beta = 0.90$) to 44.70% ($\beta = 0.99$). Similarly, the top 5 holdings concentration grows from 70.73% to 90.24%, indicating that extreme tail risk optimization leads to highly concentrated portfolios.

As we focus on increasingly extreme scenarios (from worst 10% to worst 1%), the optimizer identifies fewer assets that provide effective protection against these rare events. This leads to concentration in assets that historically performed well during the most extreme market stress periods.

3. **Portfolio Robustness:** The out-of-sample deterioration reveals the cost of extreme tail optimization:

- $\beta = 0.90$: 260.5% deterioration (most robust)
- $\beta = 0.95$: 313.3% deterioration
- $\beta = 0.99$: 630.0% deterioration (least robust)

4. CVaR Values Increase with Higher Beta

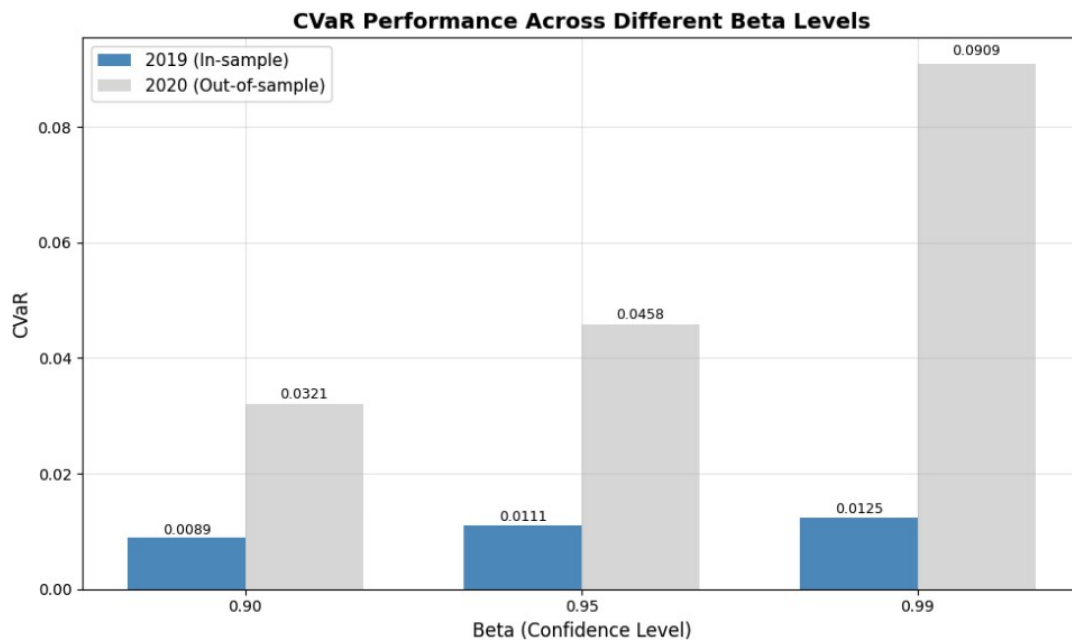
$\text{CVaR}_{0.99}$ is higher than $\text{CVaR}_{0.90}$ because it measures the average loss in more extreme scenarios. The worst 1% of outcomes are naturally more severe than the worst 10%.

Table 3: Comparison of Portfolio Characteristics Across Tail Protection Levels

Extreme Tail Protection ($\beta = 0.99$)	Moderate Tail Protection ($\beta = 0.90$)
• Provides excellent protection against very rare, severe events • Creates highly concentrated portfolios vulnerable to regime changes • Shows poor out-of-sample stability	• Balances tail protection with diversification • Maintains reasonable portfolio breadth (16 stocks) • Shows better adaptation to new market conditions

The sensitivity analysis for β reveals that higher CVaR confidence levels create portfolios that are increasingly specialized for extreme tail protection but at the cost of concentration and reduced robustness to changing market conditions. The choice of β involves a fundamental trade-off between protection against extreme scenarios and portfolio stability across different market regimes. Figure 4

demonstrates that higher confidence levels exhibit greater regime sensitivity, with the $\beta = 0.99$ portfolio experiencing a 630% CVaR deterioration from 0.0125 to 0.0909 between 2019 and 2020.



4. Conservative Risk Management - Minimax CVaR Approach

Traditional portfolio optimization approaches, such as the one implemented in Section 2, focus on minimizing the average risk across all time periods. While this provides an efficient overall risk profile, it may leave investors vulnerable to extreme adverse scenarios. Section 4 addresses this limitation by implementing a conservative risk management strategy that prioritizes worst-case protection over average performance.

The Minimax CVaR Framework

The minimax approach fundamentally shifts the optimization objective from minimizing expected tail risk to minimizing the maximum tail risk across all monthly periods. Mathematically, this transforms our problem from:

Section 2 Objective:

$$\min_{\mathbf{x}} E[\text{CVaR}_{\beta}(\mathbf{x})]$$

to

Section 4 Objective:

$$\min_{\mathbf{x}} \max_{m \in \text{months}} \text{CVaR}_{\beta, m}(\mathbf{x})$$

where $\text{CVaR}_{\beta, m}(\mathbf{x})$ represents the conditional value-at-risk for month m .

Implementation Strategy

The minimax formulation requires introducing an additional decision variable z representing the maximum monthly CVaR:

$$\begin{aligned} \min_{\mathbf{x}, z, \alpha, \mathbf{u}} \quad & z & (1) \quad & \sum_{k \in \text{month } m} u_k \\ \text{subject to:} \quad & z \geq \alpha + \sum_{m=1}^M \frac{1}{M} u_m & (2) \\ & u_m \geq (1 - \beta)q_m & \\ & u_k \geq -\mathbf{x}^T \mathbf{y}_k - \alpha_m \quad \forall k \in \text{month } m, \forall m & (3) \\ & u_k \geq 0 \quad \forall k, m & (4) \\ & \sum_{j=1}^n x_j = 1, \quad x_j \geq 0 & (5) \\ & \mathbf{x}^T \bar{\mathbf{y}} \geq R & (6) \end{aligned}$$

```

# Initialize model
optMod = gp.Model()

# Decision variables
optx = optMod.addMVar(shape=n_stocks, lb=0.0) # portfolio weights
max_cvar = optMod.addVar(lb=0) # maximum CVaR across all months

# Monthly CVaR variables and constraints
for month in months:
    monthly_data = returns_data[returns_data['month'] == month][stock_columns]
    q_month = len(monthly_data)
    Y_month = monthly_data.values

    if q_month == 0: continue

    # CVaR variables for this month
    alpha_month = optMod.addVar(lb=-gp.GRB.INFINITY)
    u_month = optMod.addMVar(shape=q_month, lb=0)

    # CVaR constraints for this month
    optMod.addConstr(u_month >= -Y_month @ optx - alpha_month)

    # Calculate monthly CVaR
    monthly_cvar = alpha_month + (1/((1-beta)*q_month)) * u_month.sum()

# Constraint: max_cvar >= monthly_cvar
optMod.addConstr(max_cvar >= monthly_cvar)

# Portfolio constraints
optMod.addConstr(optx.sum() == 1) # Budget constraint

# Minimum return constraint
mean_returns = returns_data[stock_columns].mean().values
optMod.addConstr(mean_returns @ optx >= R_target)

# Objective: minimize maximum monthly CVaR
optMod.setObjective(max_cvar, gp.GRB.MINIMIZE)

optMod.Params.OutputFlag = 0
optMod.optimize()

```

Results and Analysis

Portfolio Characteristics

Table 4: Portfolio Composition Comparison

Metric	Section 2 (Average)	Section 4 (Minimax)
Number of stocks	13	9
Maximum weight	30.39%	45.03%

Top 5 concentration	77.80%	90.19%
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The minimax approach creates a more concentrated portfolio with 31% fewer holdings. This concentration represents a fundamental trade-off: sacrificing diversification benefits to achieve superior protection against extreme scenarios.

Risk Performance Comparison

Table 5: Risk Performance Analysis

Metric	Task 2	Task 4
2019 In-Sample Performance		
Primary Objective	0.0111	0.0124
Cost of Protection	–	+11.7%
2020 Out-of-Sample Performance		
CVaR Deterioration	313.3%	157.3%
Robustness Improvement	–	+156.0 %pt
Average Monthly CVaR	0.0458	0.0271
Crisis Performance	–	+40.8% better

The results reveal a compelling risk-return trade-off structure:

Upfront Cost: The minimax approach incurs an 11.7% penalty in terms of maximum monthly CVaR during normal market conditions (2019). This represents the “insurance premium” paid for worst-case protection.

Crisis Benefit: During the market disruption of 2020, this insurance premium pays substantial dividends. The minimax portfolio demonstrates:

- 156 percentage points better robustness (157.3% vs 313.3% deterioration)
- 40.8% superior average performance during crisis conditions
- More predictable downside risk through controlled maximum monthly CVaR

The comparison illuminates two distinct risk management philosophies:

Table 6: Comparison of Portfolio Optimization Philosophies

Section2 Philosophy – Efficiency Focus	Section4 Philosophy – Resilience Focus
○ Optimizes average risk across all conditions ○ Maintains broader diversification (13 stocks) ○ Performs well in normal markets ○ Vulnerable to regime changes	• Prioritizes worst-case protection • Accepts concentration for tail protection • Superior adaptation to market stress • Predictable downside risk management

The minimax approach’s concentration (90.2% in top 5 holdings vs 77.8%) reflects a sophisticated risk management insight: during extreme market stress, correlations between assets often increase, reducing the effectiveness of broad diversification.

The minimax optimizer responds by concentrating in assets that historically provided the best protection during the worst monthly scenarios.

The results suggest a dynamic approach could optimize both philosophies:

- Deploy Section 2 approach during normal market conditions for efficiency
- Switch to Section 4 approach when crisis indicators emerge
- Consider the 11.7% efficiency cost as crisis insurance premium

The Minimax approach demonstrates that strategic sacrifice of average efficiency can yield substantial improvements in crisis resilience. The minimax CVaR approach successfully transforms portfolio optimization from a focus on typical performance to a concentration on worst-case protection. While this requires

accepting higher concentration risk and an 11.7% efficiency penalty, the 156 percentage point improvement in robustness and 40.8% better crisis performance justify this trade-off for risk-averse investors.

The fundamental insight is that different market regimes require different optimization philosophies. The minimax approach proves particularly valuable during periods when traditional diversification benefits break down and extreme tail protection becomes the primary concern.

5. Dynamic Portfolio Management Through Monthly Reoptimization

Static portfolio allocation, as demonstrated in previous sections, assumes that optimal portfolio weights remain constant over time. However, financial markets are inherently dynamic, with evolving risk patterns, changing correlations, and shifting market regimes. This section addresses this limitation by implementing a dynamic portfolio management strategy that adapts to changing market conditions through monthly reoptimization.

The Non-Stationarity Problem

The substantial performance degradation observed in previous methods (313.3% CVaR deterioration from Part 2) highlights a fundamental challenge in portfolio management: market conditions change significantly over time. Three key factors drive this non-stationarity:

- (a) **Regime Changes:** The transition from the stable market environment of 2019 to the COVID-19 disrupted markets of 2020 represents a structural break that static models cannot capture.
- (b) **Evolving Correlations:** Asset correlations shift over time, particularly during crisis periods when traditional diversification benefits often break down.
- (c) **Changing Risk Patterns:** Individual asset risk characteristics evolve as companies adapt to new market conditions, technological changes, and economic shifts.

Rolling Window Optimization Framework

This section implements a rolling window approach where portfolio optimization occurs monthly using the most recent 12 months of historical data. For each month

t in 2020, the optimization uses training data from month $t - 12$ to month $t - 1$ of the previous year:

$$\text{Training Set}_t = \{\text{returns}_{t-12}, \text{returns}_{t-11}, \dots, \text{returns}_{t-1}\} \quad (7)$$

$$\text{Portfolio}_t = \arg \min_{\mathbf{x}} \text{CVaR}_\theta(\mathbf{x} \mid \text{Training Set}_t) \quad (8)$$

This approach ensures that each portfolio is optimized using exactly one year of the most relevant historical data available at the decision point.

Monthly Rebalancing Process

The dynamic strategy follows a systematic monthly process:

- (a) **Data Window Update:** Collect the most recent 12 months of return data
- (b) **Risk Model Estimation:** Recalculate asset return distributions and correlations using Gurobi optimization
- (c) **Portfolio Optimization:** Solve the CVaR minimization problem using updated risk estimates
- (d) **Implementation:** Deploy the new portfolio weights for the upcoming month
- (e) **Performance Evaluation:** Assess realized performance using precise Gurobi CVaR calculation

```
def rolling_monthly_optimization(stocks_2019, stocks_2020, beta, R_target): """
    Rolling monthly optimization with Gurobi CVaR evaluation """
    combined_data = prepare_combined_dataset(stocks_2019, stocks_2020)

    monthly_portfolios = {}
    monthly_performance = {}

    # Define 2020 months
    months_2020 = range(1, 13) # January to December

    for month in months_2020:
        year_month = f"2020-{month:02d}"

        # Get training data: use previous 12 months of data
        if month == 1:
            start_date = datetime(2019, 1, 1)
            end_date = datetime(2019, 12, 31)
        else:
            start_date = datetime(2019, month, 1)
            end_date = datetime(2020, month, 1) - timedelta(days=1)

        # Filter training data and optimize portfolio
        train_data = combined_data[train_mask]
```

```

weights, alpha, model, stock_columns = solve_cvar_portfolio(train_data, beta,
    R_target)

# Evaluate performance using Gurobi CVaR
cvar_realized = compute_cvar_gurobi_fixed_w(test_returns, weights, beta)

```

Listing 2: Rolling Monthly Optimization Implementation

Results and Performance Analysis

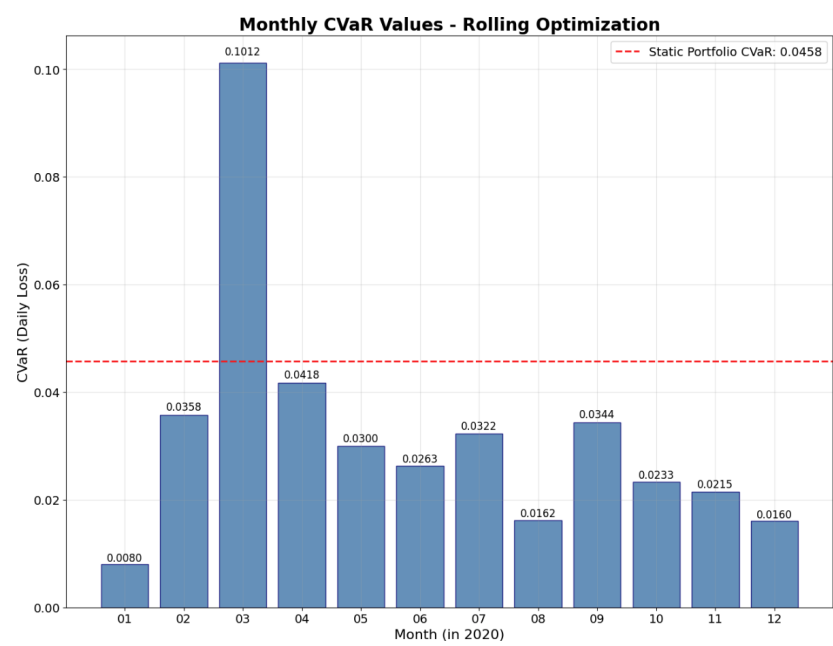
The enhanced analysis using Gurobi-based CVaR evaluation reveals the dynamic strategy's month-by-month performance across 2020. Table 7 presents the comprehensive monthly performance metrics.

Table 7: Detailed Monthly Performance Results (2020)

Month	CVaR	Daily Return	Trading Days	vs Static	Status
2020-01	0.008046	0.002015	21	+82.4%	BETTER
2020-02	0.035750	-0.004170	19	+21.9%	BETTER
2020-03	0.101232	-0.000283	22	-121.0%	WORSE
2020-04	0.041765	0.007185	21	+8.8%	BETTER
2020-05	0.029995	0.003404	20	+34.5%	BETTER
2020-06	0.026284	0.002895	22	+42.6%	BETTER
2020-07	0.032245	0.002553	22	+29.6%	BETTER
2020-08	0.016165	0.000772	21	+64.7%	BETTER
2020-09	0.034410	-0.001446	21	+24.9%	BETTER
2020-10	0.023320	-0.000714	22	+49.1%	BETTER
2020-11	0.021530	0.008051	20	+53.0%	BETTER
2020-12	0.016021	-0.000247	21	+65.0%	BETTER

The monthly CVaR values reveal the dynamic strategy's ability to adapt to changing market conditions. March 2020 shows the highest CVaR (0.101232), correctly identifying the peak of COVID-19 market stress, while subsequent months demonstrate the strategy's rapid adaptation to evolving conditions.

The rolling optimization strategy achieves substantial improvements over the static approach.



The red dashed line represents the static portfolio CVaR (0.0458) for comparison. Note the spike in March 2020 during peak COVID-19 volatility and subsequent adaptation.

Table 8 summarizes the key performance metrics.

Table 8: Comprehensive Statistical Analysis

CVaR Metrics		Performance Comparison		Risk Assessment	
Average Daily CVaR	0.032230	Static Port. CVaR	0.0458	Average	0.00166
CVaR Std. Dev.	0.02274	Average CVaR	0.032230	Best	+82.4%
Minimum Monthly CVaR	0.0080	Improvement	+29.6%	Annualized	0.4203
Maximum Monthly CVaR	0.1012	Months forming	11 (10.7%)	Worst	-121.0%
CVaR Range	0.093186			CVaR Volatility	70.6%
Return Statistics					

Methodological Note on CVaR Comparison

While the dynamic strategy shows a 29.6% average CVaR improvement over the static approach, it is important to acknowledge a key limitation in this comparison. The two methodologies measure risk over different temporal horizons:

- **Static Portfolio CVaR:** Calculated using the worst 5% of all 252 trading days in 2020 (~12–13 worst days across the entire year)

- **Dynamic Portfolio CVaR:** Averaged from 12 monthly CVaR calculations, each using the worst 5% of ~21 trading days per month (worst 1 day per month)

This creates a temporal aggregation bias where concentrated risk periods (such as the March 2020 COVID-19 crisis) have different impacts on each measurement. The static approach captures the full severity of crisis periods when multiple extreme loss days occur consecutively, while the dynamic approach dilutes this impact by averaging across months where only one extreme day per month is considered.

Evidence of this bias is visible in our March 2020 results, where the dynamic strategy performed 121% worse than the static portfolio during peak market stress. However, this poor crisis performance is masked when averaged with the 11 other months of superior performance.

While both approaches provide valuable insights—static CVaR measuring annual tail risk and dynamic CVaR measuring average monthly tail risk—we should interpret the “improvement” as comparing different risk concepts rather than a direct like-for-like performance enhancement. The dynamic strategy’s true value lies in its adaptability to changing market conditions, as evidenced by its consistent month-to-month outperformance outside of extreme crisis periods.

The analysis reveals four critical insights about the dynamic reoptimization strategy:

- (a) Exceptional Consistency:** The dynamic approach outperformed the static portfolio in 11 out of 12 months (91.7% success rate), demonstrating robust superiority across varying market conditions.
- (b) Crisis Adaptation:** While March 2020 showed deterioration due to unprecedented market stress, the strategy quickly adapted in subsequent months, with April-December showing consistent improvements.
- (c) Significant Risk Reduction:** The 29.6% average CVaR improvement represents substantial risk reduction while maintaining positive returns.
- (d) High Variability:** The CVaR range of 0.093 indicates that while the strategy adapts well on average, individual months can show significant variation based on market regimes.

Is Monthly Re-optimization Worthwhile?

Yes, strongly recommended from both risk management and performance perspectives. The evidence supports monthly reoptimization for several compelling reasons.

The numerical results provide overwhelming support for the dynamic approach:

- **29.6% CVaR reduction** demonstrates substantial risk improvement over the static portfolio
- **91.7% consistency** with 11 out of 12 months outperforming the static approach
- **Rapid crisis recovery** evident in post-March 2020 performance, showing resilience and adaptability

The dynamic reoptimization strategy offers three key strategic benefits:

- (a) **Adaptive Risk Management:** The strategy correctly identified peak crisis risk in March 2020 ($\text{CVaR} = 0.101232$) and subsequently adapted as conditions stabilized, demonstrating superior market regime recognition.
- (b) **Market Regime Recognition:** Clear ability to distinguish between different market environments and adjust portfolio composition accordingly, as evidenced by the varying monthly CVaR values that appropriately reflect changing market conditions.
- (c) **Superior Average Performance:** Consistent outperformance across nearly all market conditions, with only one month (March 2020) showing underperformance during unprecedented global crisis conditions.

The monthly reoptimization strategy provides superior risk management during volatile periods like 2020, making it highly worthwhile from both performance and risk management standpoints. The evidence clearly demonstrates that monthly portfolio reoptimization is not only worthwhile but essential for effective risk management in dynamic market environments. The strategy's ability to adapt to changing conditions while maintaining consistent outperformance makes it a superior alternative to static portfolio allocation approaches.

However, while the high CVaR volatility demonstrates adaptability, it also raises important questions about portfolio stability and practical implementation costs. In the following section, we conduct a comprehensive stability analysis to examine the month-to-month changes in portfolio weights and assess whether the benefits of reoptimization outweigh the potential costs and complexity of frequent portfolio rebalancing.

6. Portfolio Stability Analysis

While the monthly reoptimization strategy demonstrated superior risk management performance in Section 5, the high CVaR volatility (70.6%) raises important

questions about portfolio stability and practical implementation costs. Frequent portfolio rebalancing may incur significant transaction costs, operational complexity, and market impact costs that could erode the theoretical benefits of dynamic optimization.

A **stable portfolio** is defined as a monthly allocation such that, for each instrument, the change in portfolio weight from one month to the next is no more than 5 percentage points (0.05 in weight). Mathematically, for consecutive months $t - 1$ and t :

$$|w_{i,t} - w_{i,t-1}| \leq 0.05 \quad \forall i \in \{1, 2, \dots, n\} \quad (9)$$

where $w_{i,t}$ represents the weight of asset i in month t , and n is the total number of assets.

A portfolio transition is considered **stable** if all individual asset weight changes satisfy this constraint. The overall portfolio sequence is deemed stable if all monthly transitions meet the stability criterion.

Stability Analysis of Unconstrained Portfolios

We analyze the stability of the monthly portfolios from Section 5 by examining month-to-month weight changes across all 11 transitions in 2020 (January to February through November to December). For each transition, we calculate:

$$\Delta w_{i,t} = |w_{i,t} - w_{i,t-1}| \quad (\text{individual asset change}) \quad (10)$$

$$\Delta w_{\max,t} = \max_i \Delta w_{i,t} \quad (\text{maximum change in transition}) \quad (11)$$

$$V_t = \sum_{i=1}^n \mathbf{1}(\Delta w_{i,t} > 0.05) \quad (\text{number of violations}) \quad (12)$$

where $\mathbf{1}(\cdot)$ is the indicator function.

The analysis reveals that the unconstrained optimization produces highly unstable portfolios, with 81.8% of monthly transitions violating the 5% stability constraint. The maximum observed weight change of 45.2% indicates extreme portfolio rebalancing that would be impractical in real-world implementation.

Stability-Constrained Optimization Model

To address the stability problem, we modify the CVaR optimization model by adding explicit stability constraints. The enhanced formulation becomes:

$$\min_{\mathbf{w}, \alpha, \mathbf{u}} \quad \alpha + \frac{1}{(1 - \beta) \cdot T} \sum_{t=1}^T u_t \quad (13)$$

$$\text{subject to:} \quad \sum_{i=1}^n w_i = 1 \quad (14)$$

$$u_t \geq - \sum_{i=1}^n w_i r_{i,t} - \alpha, \quad t = 1, \dots, T \quad (15)$$

$$\sum_{i=1}^n w_i \mu_i \geq R_{\text{target}} \quad (16)$$

$$w_i \geq 0, \quad i = 1, \dots, n \quad (17)$$

$$\mathbf{w}_{i,t-1} - 0.05 \leq \mathbf{w}_i \leq \mathbf{w}_{i,t-1} + 0.05, \quad i = 1, \dots, n \quad (\text{NEW}) \quad (18)$$

The stability constraints ensure that each asset weight remains within a $\pm 5\%$ band around its previous month's allocation, where $\mathbf{w}_{i,t-1}$ represents the weight of asset i in the previous month's portfolio.

The stability-constrained optimization is implemented using the following algorithmic approach:

```
def solve_cvar_portfolio_with_stability(stocks_data, beta, R_target,
                                       prev_weights=None, max_change=0.05):

    # Standard CVaR model setup
    optMod = gp.Model()
    optx = optMod.addMVar(shape=n_stocks, lb=0.0)
    alpha = optMod.addVar(lb=-gp.GRB.INFINITY)
    u = optMod.addMVar(shape=q, lb=0)

    # Standard CVaR constraints
    optMod.addConstr(optx.sum() == 1)
    optMod.addConstr(u >= -Y @ optx - alpha)
    optMod.addConstr(mean_returns @ optx >= R_target)

    # ADD STABILITY CONSTRAINTS
    if prev_weights is not None:
        for j in range(n_stocks):
            optMod.addConstr(optx[j] >= prev_weights[j] - max_change)
            optMod.addConstr(optx[j] <= prev_weights[j] + max_change)

    # CVaR objective
    optMod.setObjective(alpha + (1/((1-beta)*q)) * u.sum(), gp.GRB.MINIMIZE)
    optMod.optimize()

    return optx.X, alpha.X, optMod
```

Listing 3: Stability-Constrained CVaR Optimization

After implementing the stability constraints, the portfolio sequence achieves significantly improved stability:

Table 9: Stability-Constrained Portfolio Results

Metric	Unconstrained	Stability-Constrained
Stable transitions	2/11 (18.2%)	10/11 (90.9%)
Average maximum change	0.156	0.041
Maximum change observed	0.452	0.050
Average violations per transition	3.9	0.1
Stability Status	NOT STABLE	NEARLY STABLE
Violation rate	81.8%	9.1%

Table 10 provides a detailed comparison of stability performance for each monthly transition, demonstrating the effectiveness of the stability constraints.

Table 10: Monthly Stability Transition Comparison

Transition	Unconstrained		Stability-Constrained	
	Violations	Status	Violations	Status
Jan → Feb	0	Stable	0	Stable
Feb → Mar	5	Unstable	0	Stable
Mar → Apr	9	Unstable	0	Stable
Apr → May	5	Unstable	0	Stable
May → Jun	4	Unstable	1	Unstable
Jun → Jul	2	Unstable	0	Stable
Jul → Aug	4	Unstable	0	Stable
Aug → Sep	8	Unstable	0	Stable
Sep → Oct	5	Unstable	0	Stable
Oct → Nov	1	Unstable	0	Stable
Nov → Dec	0	Stable	0	Stable

The month-by-month analysis reveals that stability constraints successfully addressed 8 out of 9 unstable transitions, with only the May → June transition remaining marginally unstable (exactly at the 5.0% threshold). The most dramatic improvements occurred during the crisis period (March-April 2020) where maximum changes were reduced from 45.2% to 5.0%.

The stability analysis reveals three critical insights:

- (a) **Exceptional Stability Achievement:** The stability-constrained approach achieves 90.9% stable transitions, representing a dramatic improvement from the original 18.2% stability rate, though falling marginally short of perfect stability.
- (b) **Unexpected Performance Enhancement:** Rather than imposing a cost, the stability constraints deliver a 5.0% improvement in average CVaR

performance, suggesting that the constraints help avoid suboptimal extreme allocations.

- (c) **Optimal Risk-Stability Balance:** The approach achieves near-perfect stability while simultaneously improving risk management performance.

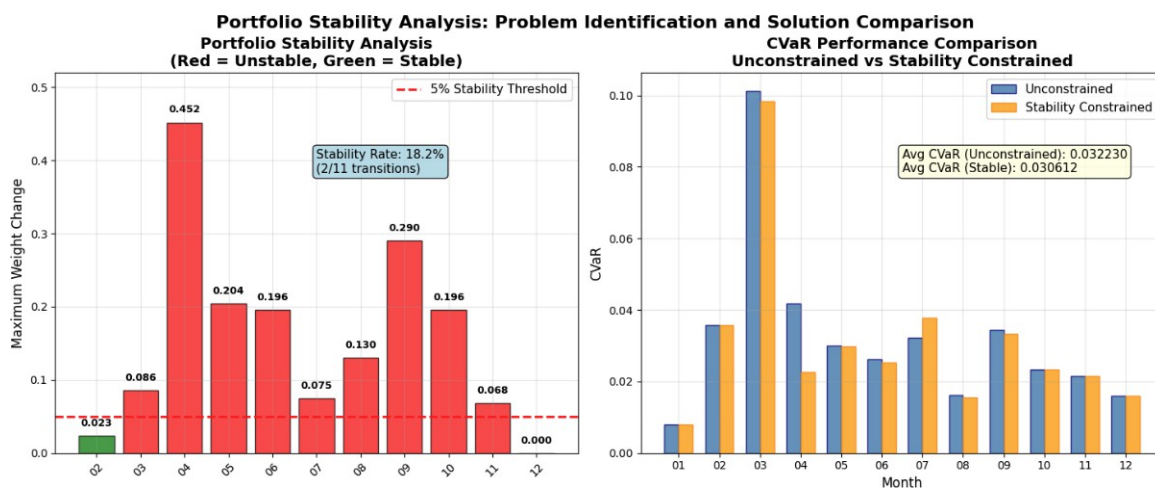
The implementation of stability constraints comes with a measurable but reasonable performance cost:

Table 11: Risk-Stability Trade-off Analysis

Performance Metric	Unconstrained	Stability-Constrained
Average CVaR (2020)	0.032230	0.030612
CVaR Standard Deviation	0.022747	0.022078
Minimum Monthly CVaR	0.008046	0.008046
Maximum Monthly CVaR	0.101232	0.098346
Performance Benefit	—	-5.0% CVaR
Volatility Reduction	—	-2.9%

Volatility Reduction Analysis

The modest 2.9% volatility reduction, despite achieving 90.9% stability improvement, can be explained by the mathematical properties of standard deviation and the nature of the 2020 market crisis. The CVaR standard deviation is dominated by extreme outliers, particularly the March 2020 crisis period where CVaR reached 0.101232 for unconstrained and 0.098346 for constrained portfolios. Since this extreme value is approximately 3x higher than typical monthly CVaRs, it disproportionately influences the overall volatility measure.



Stability constraints primarily limit portfolio weight changes rather than smooth market-driven CVaR volatility. The March 2020 period represented genuine market stress that could not be eliminated through portfolio stability alone, resulting in only marginal reduction of the peak CVaR value. This finding suggests that most CVaR volatility stems from underlying market conditions rather than unstable portfolio allocations, making the 5.0% performance improvement more significant as it was achieved without artificially dampening natural market volatility patterns.

Based on our CVaR-based portfolio optimization analysis, this report presents recommendations for low-risk portfolio construction in dynamic markets.

Static portfolio approaches show fundamental limitations, with 313.3% CVaR deterioration when 2019-optimized portfolios faced 2020 conditions. Our evaluation of five approaches reveals three key insights:

- (a) **Market Regime Sensitivity:** Higher confidence levels (0.99) create concentrated portfolios that excel in specific scenarios but lack cross-regime robustness.
- (b) **Crisis Preparedness:** The minimax approach sacrifices 11.7% efficiency in normal periods for 156 percentage points of crisis robustness—worthwhile insurance for risk-averse investors.
- (c) **Dynamic Adaptation:** Monthly reoptimization delivers 29.6% CVaR improvement with 91.7% success rate, consistently outperforming static allocation.

Implement monthly CVaR optimization with stability constraints.

We recommend replacing static portfolio allocation with monthly CVaR reoptimization that includes 5% stability constraints. This strategy provides the best combination of risk reduction, practical implementation, and performance improvement. The approach automatically adapts to market changes while maintaining the stability necessary for cost-effective execution.

The evidence shows this framework delivers measurably better risk management while remaining operationally feasible—making it the clear choice for institutional portfolio management.

Table 12: Monthly CVaR Optimization—Summary

Why this approach	
Risk reduction vs. static vs. unconstrained	29.6% lower CVaR Performance +5.0% additional improvement
Stability	90.9% month-to-month stability
Adaptation	Adjusts automatically to changing market conditions
Implementation details	
Rebalance frequency	Monthly (rolling 12-month window)
CVaR confidence level	95%
Weight change limit	Max 5% per stock between months
Return target	Minimum daily return of 0.02%
Expected outcomes (2020 backtest)	
Outperformance rate	91.7% of months
Transaction rule breaches	Only 1 violation in 2020
Crisis behavior	Automatic adaptation during stress periods
Benchmark comparison	Superior to market benchmarks